

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

SUBMMITED TO:

MR. RASHAN RAJ

SUBMMITED BY:

MANKIRAT SINGH

ID: BCS2024542

ROLL NO: 2410303257

Lab Assignment 8: Classes and Objects

Experiment No. 8.1 - Class Student with one object

Experiment Title:

Class Student with one object

Aim:

To create a class Student and one object, and call its show() method.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

A class is a blueprint that describes the fields (data) and methods (behaviour) of a kind of object. An object is a real instance created from the class with the new keyword. Many objects can be created from one class, and each has its own copy of the fields. A method with return type boolean returns true or false.

Algorithm / Procedure:

1. Define class Student with fields roll and name and method show().
2. In main, create an object with new Student().
3. Set roll = 5 and name = Rohan.
4. Call show().

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 8, Task 1
// Question: Create class Student with int roll, String name and
// method show() that prints "Roll:" and the roll, then "Name:" and
// the name. In main, create a Student with roll 5 and name Rohan and
// call show().

class Student {
    int roll;
    String name;

    void show() {
        System.out.println("Roll: " + roll);
        System.out.println("Name: " + name);
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Student s = new Student();
        s.roll = 5;
        s.name = "Rohan";
        s.show();
    }
}
```

Input:

Roll = 5, Name = Rohan.

Output:

```
Roll: 5
Name: Rohan
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. The Student object was created and its details were displayed.

Viva-Voce Questions:

- What is the difference between a class and an object?
- What does the new keyword do?
- What is a field?

Experiment No. 8.2 - Class Student with two objects

Experiment Title:

Class Student with two objects

Aim:

To create two independent objects of the same class.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

A class is a blueprint that describes the fields (data) and methods (behaviour) of a kind of object. An object is a real instance created from the class with the new keyword. Many objects can be created from one class, and each has its own copy of the fields. A method with return type boolean returns true or false.

Algorithm / Procedure:

1. Define class Student with fields roll and name and method show().
2. Create objects s1 and s2 with new Student().
3. Set s1 to (5, Rohan) and s2 to (8, Meera).
4. Call show() on each object.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 8, Task 2
// Question: Create class Student with int roll, String name and
// method show() that prints "Roll:" and the roll, then "Name:" and
// the name. Create two Students: roll 5 named Rohan, and roll 8
// named Meera. Call show() on each.

class Student {
    int roll;
    String name;

    void show() {
        System.out.println("Roll: " + roll);
        System.out.println("Name: " + name);
    }
}

public class Main {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.roll = 5;
        s1.name = "Rohan";

        Student s2 = new Student();
```

```
        s2.roll = 8;
        s2.name = "Meera";

        s1.show();
        s2.show();
    }
}
```

Input:

Student 1: Roll 5, Rohan. Student 2: Roll 8, Meera.

Output:

```
Roll: 5
Name: Rohan
Roll: 8
Name: Meera
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. Two objects were created and each kept its own field values.

Viva-Voce Questions:

- Does changing s1 affect s2?
- How many objects can one class create?
- What is a reference variable?

Experiment No. 8.3 - Class Book with isLong()

Experiment Title:

Class Book with isLong()

Aim:

To write a method that returns a boolean based on a field value.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

A class is a blueprint that describes the fields (data) and methods (behaviour) of a kind of object. An object is a real instance created from the class with the new keyword. Many objects can be created from one class, and each has its own copy of the fields. A method with return type boolean returns true or false.

Algorithm / Procedure:

1. Define class Book with field pages.
2. Write boolean isLong() that returns pages > 300.
3. Create two Book objects with 450 and 120 pages.
4. Print isLong() of each object.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 8, Task 3
// Question: Create class Book with int pages and method isLong()
// that returns true when pages is more than 300. In main, create a
// Book with 450 pages and one with 120 pages, and print isLong() for
// each, one per line.

class Book {
    int pages;

    boolean isLong() {
        return pages > 300;
    }
}

public class Main {
    public static void main(String[] args) {
        Book b1 = new Book();
        b1.pages = 450;

        Book b2 = new Book();
        b2.pages = 120;

        System.out.println(b1.isLong());
```

```
        System.out.println(b2.isLong());
    }
}
```

Input:

Book 1: 450 pages. Book 2: 120 pages.

Output:

```
true
false
```

Time / Space Complexity: Time O(1); Space O(1).

Result:

The program was compiled and executed successfully. isLong() returned true for 450 pages and false for 120 pages.

Viva-Voce Questions:

- What does a boolean method return?
- How does return pages > 300 work?
- Can two objects give different results from the same method?